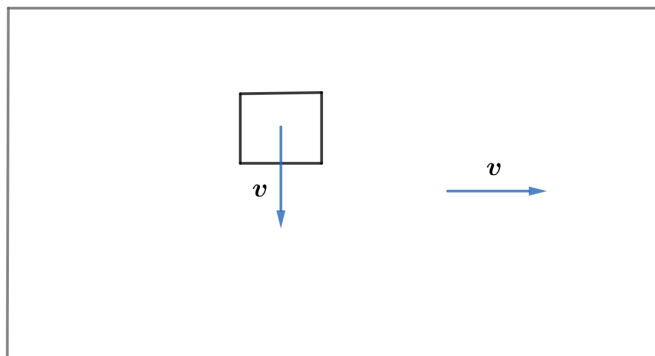


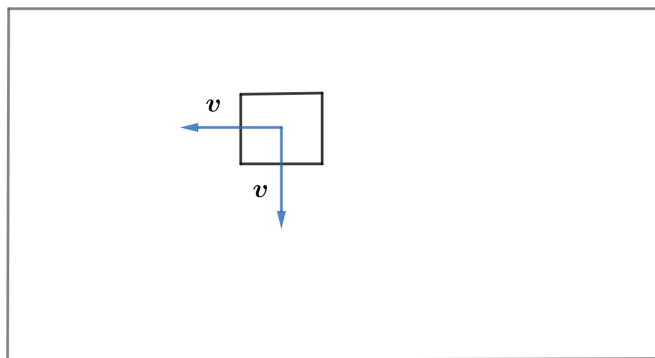
2020B F=ma Exam: Problem 8

Kevin S. Huang

Ground frame:



Belt frame:



In the belt frame, the box has speed $\sqrt{2}v$. Since

$$a = \frac{f}{m} = \frac{\mu mg}{m} = \mu g$$

we have from kinematics,

$$v\sqrt{2} - \mu g t = 0$$

$$t = \frac{v\sqrt{2}}{\mu g}$$

so the answer is .